

Modeling Knowledge

Yesterday

Today

Tomorrow



Peter Edelsbrunner
Erfurt 2026

<https://github.com/peter1328/presentations>

Obviously, children's conceptual knowledge is structured like a coherent theory about a topic!



S. Vosniadou

You are both right and wrong! It differs between children – there is heterogeneity!



M. Schneider

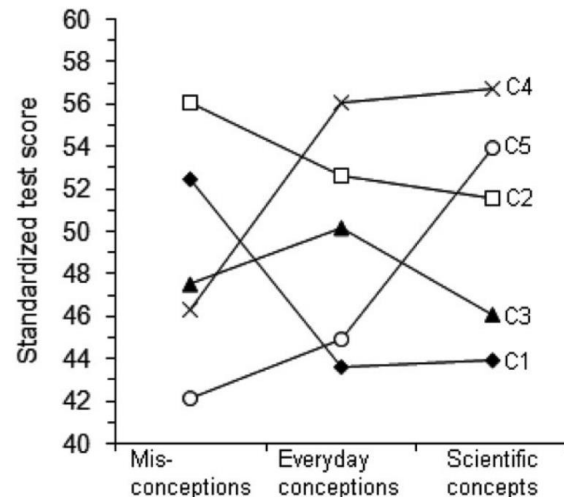
Stella, you are wrong! Children's knowledge is not theory-like, it consists of many loose intuitions!



A. diSessa

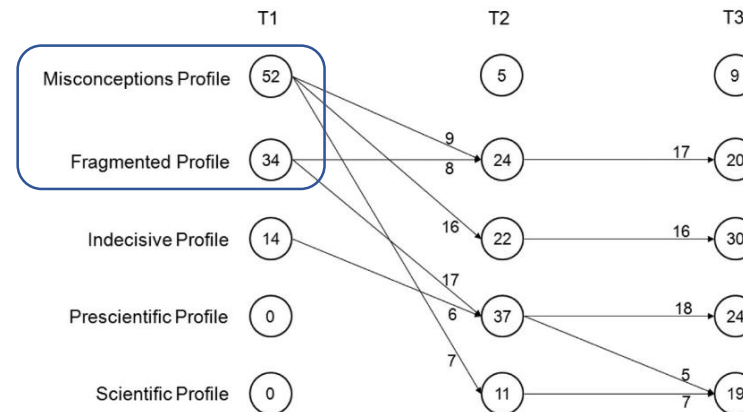
Schneider & Hardy (2013)

Latent transition analysis



Mixture modeling

Children develop through knowledge states (e.g., fragmented -> coherent)



Theory

Model

Part I

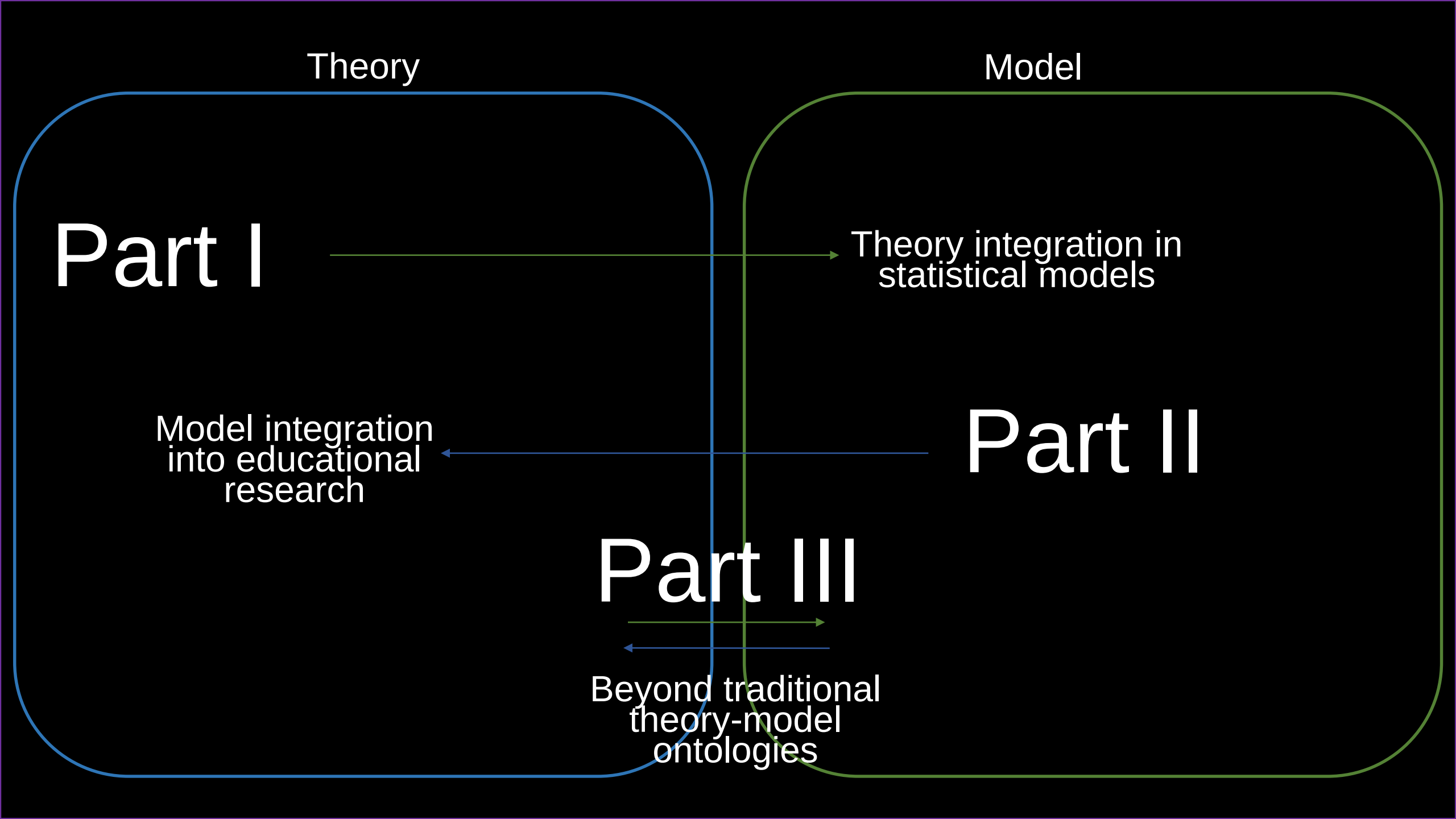
Theory integration in
statistical models

Model integration
into educational
research

Part II

Part III

Beyond traditional
theory-model
ontologies



Theory

Model

Part I

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Theory

70s/80s

Model

Inhelder & Piaget's Theory of stage-wise development (1958)

Siegler's (1976) ruled-based assessment methodology

Sato's (1975) erroneous rules/bugs identification approach

Example subtraction items:

-16 - (-7)	000
2 - 11	001
12 + (-3)	010
	100
	011
	101
	110
	111

Kikumi Tatsuoka (1983)

I've got an awesome idea!

Mapping:
Solution pattern $\leftrightarrow$ cognitive rule
(i.e., solution strategy)



e.g.

„011“ $\rightarrow$ 80% prob. produced by misconception („bug“) leading you to subtract -7 from -16

Latent variable model (cf. *cognitive diagnosis model*):

- Bugs (i.e., misconceptions) as **latent mental states**
- Each response pattern produced by each bug with specific probability
- Emphasis on **patterns across multiple items**
- **Different strategies/misconceptions** can lead to **same (sum) score**

Theory

70s/80s

Model

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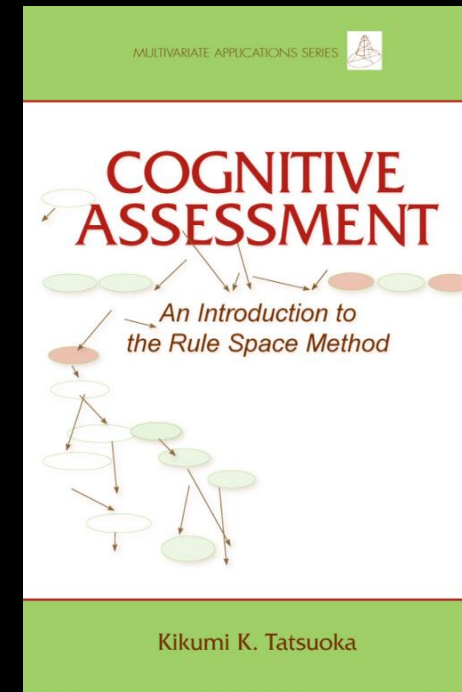
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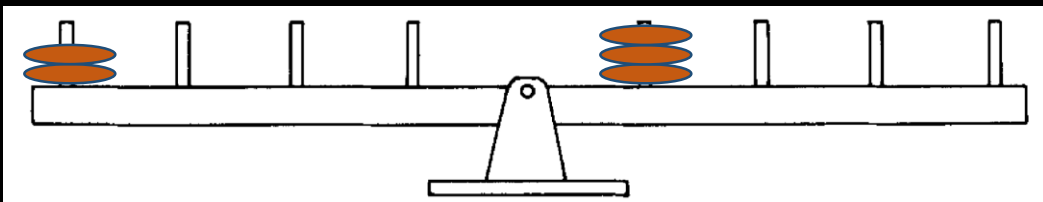
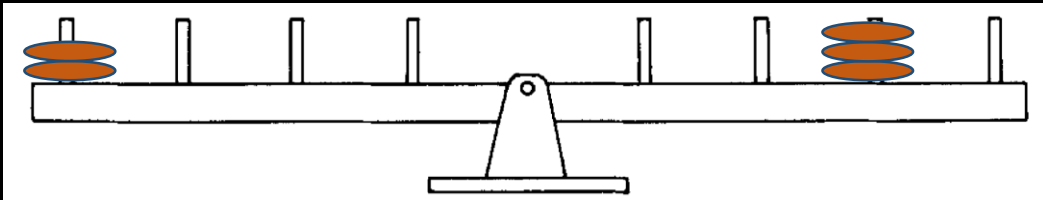
70s/80s

Model

Inhelder & Piaget's Theory of stage-wise development (1958)

Siegler's (1976) ruled-based assessment methodology

Consider (1) weight, (2) distance, (3) proportion of w/d



Mark Wilson (1985/1989)



This is so cool,
let's model it!

Extension of the **Rasch model**

Items have different difficulty but it depends on a person characteristic:

The cognitive rule (i.e., developmental stage)

Rule I (one-dimensional reasoning, e.g. weight)

...

Rule IV (proportional reasoning distance x weight)

Depending on rule person uses, specific items become easier (or harder)

Theory

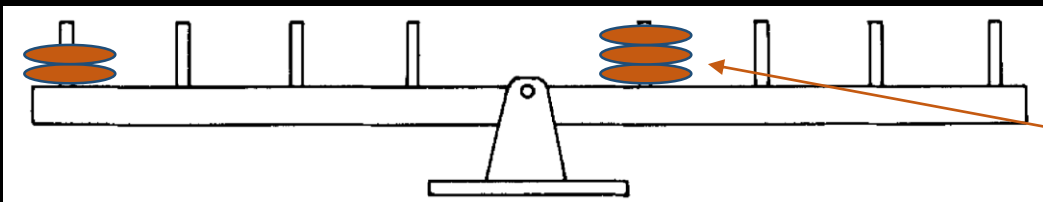
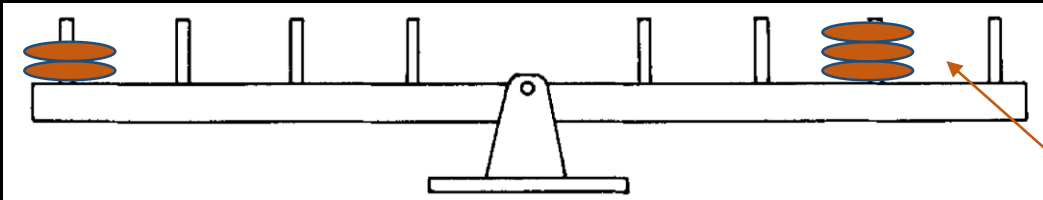
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Extension of the **Rasch model**

The **Saltus model**

Each **response pattern** indicates the **probability** for each of the four latent **mental states** (Rule I – IV)

Item types	RI	RII	RIII	RIV
Balance	1.00	1.00	1.00	1.00
Weight	1.00	1.00	1.00	1.00
Distance	.00	1.00	1.00	1.00
Conflict-balance	.00	.00	.33	1.00
Conflict-weight	1.00	1.00	.33	1.00
Conflict-distance	.00	.00	.33	1.00

$$\sigma_{ij} = \theta_i - \beta_j + \sum_h \phi_{ih} \tau_{hk}$$

Theory

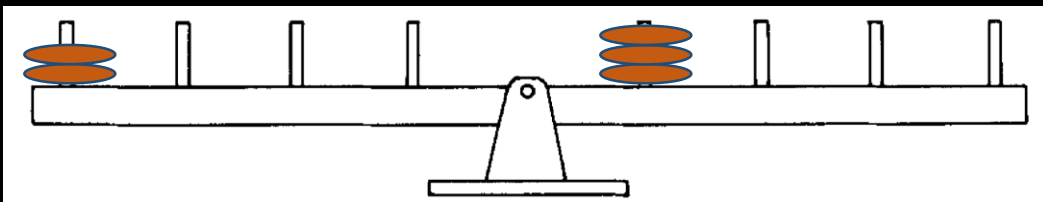
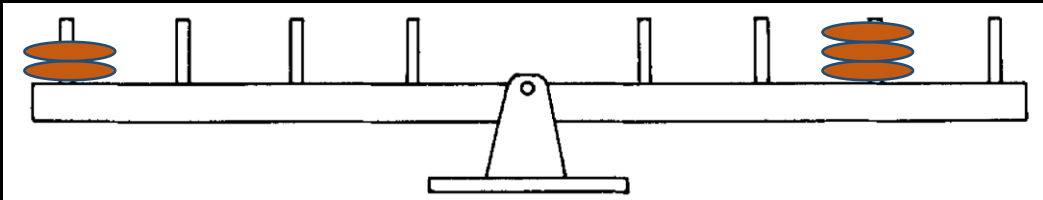
70s/80s

Model

Inhelder & Piaget's Theory of stage-wise development (1958)

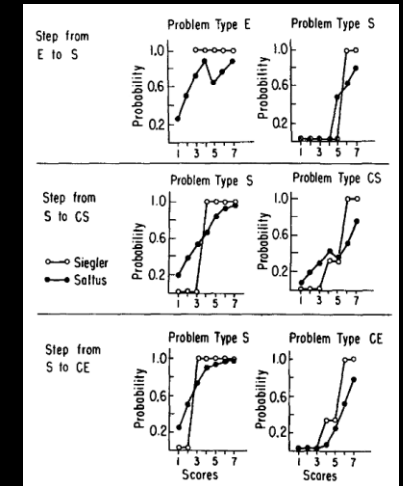
Siegler's (1976) ruled-based assessment methodology

Consider (1) weight, (2) distance, (3) proportion of w/d



Mark Wilson (1985/1989)

Model showed **deviations** from some of Siegler's **predictions**



- Difference between **person ability** and **item difficulty** determines solution probability
- **Bonus/malus on solution probability** depending on latent rule state
- Latent rule h differs across persons, bonus/malus τ across item groups k

$$\sigma_{ij} = \theta_i - \beta_j + \sum_h \phi_{ih} \tau_{hk}$$

Theory

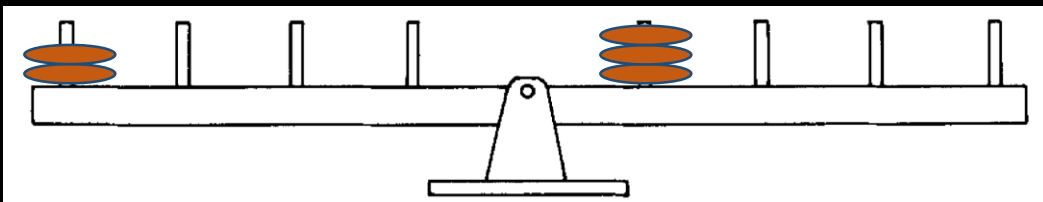
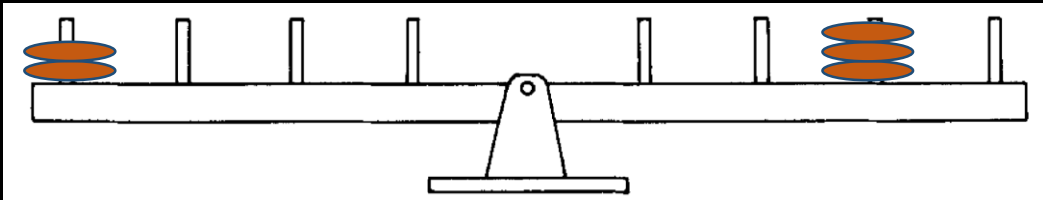
90s/2000s

Model

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Consider (1) weight, (2) distance, (3) proportion of w/d



Jansen & van der Maas (1997, 2002)



Constrained latent class analysis

Formalized further potential rules:

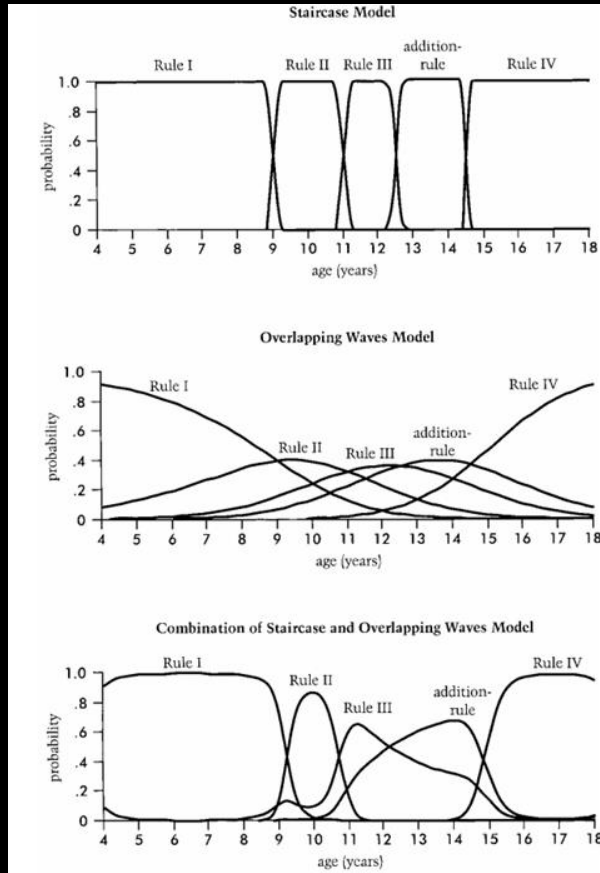
- Random answer in case of balance
- Balance in case of w/d conflict
- Acknowledgment of *large* distances only
- Restriction of known item solution probabilities for Siegler Rules I – IV

Theory

90s/2000s

Model

Result: Integrative theoretical *model of reasoning about the balance scale task*



Jansen & van der Maas (1997, 2002)



Constrained latent class analysis

Results

$N = 484$ 7th- and 8th-graders:
Rules I and II very consistent, rules above partially

Further applications:

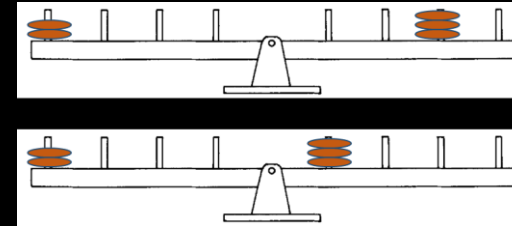
- Rejection of connectonist models as rule-behaving
- Modeling of rules in transition phases, differences across age and development

Kikumi Tatsuoka (1983)

Mark Wilson (1985/1989)

Jansen & van der Maas (1997, 2002)

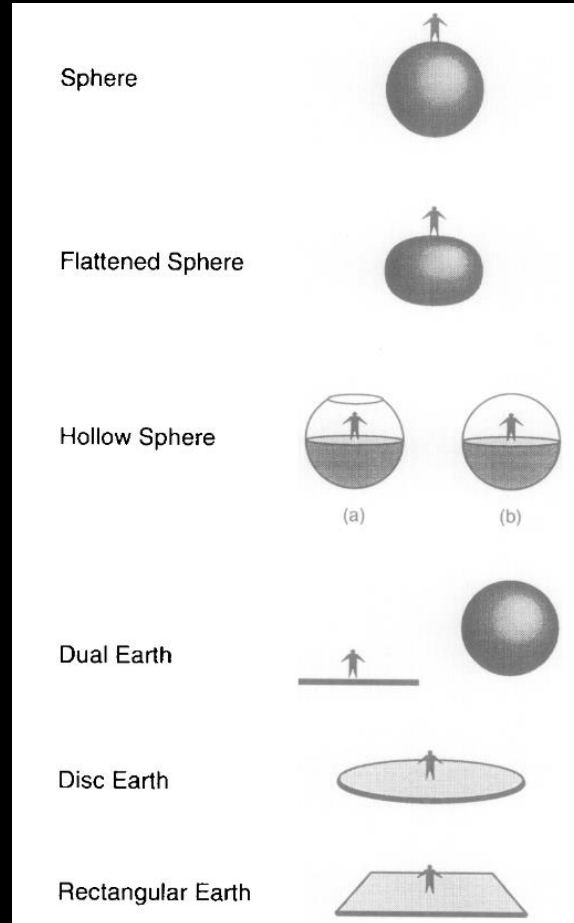
Modeling theoretical assumptions allowed:



- Determining **probabilistic** misconception/rule use for each child
- Estimating and comparing proportions of children showing each misconception/rule
- Estimating effects of practice/interventions on misconceptions/rule use
- Adaptive instruction (cf. Intelligent tutoring systems)
- Comparing theoretical predictions with empirical model estimates; theory development

Theory Conceptual Change Model

Vosniadou & Brewer (1992)



Straatemeier, Jansen, & van der Maas (2008)



- Children are typically classified according to answers in interviews, drawing, or claying tasks
- Arbitrary classification cut-offs (e.g., > 80% in accordance with one predicted pattern), lack of possibility to identify unexpected patterns, no possibility to test goodness-of-fit (how well does our classification account for the observed data?)
- Constrained latent class analysis using a new pen-and-paper test (EARTH-2; 8 items + Interviews)
- Kindergarteners – 3rd grade

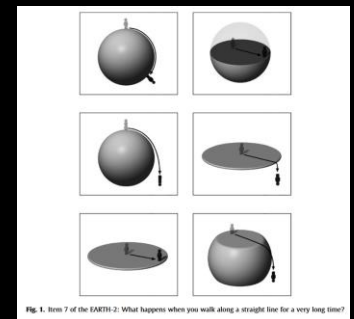
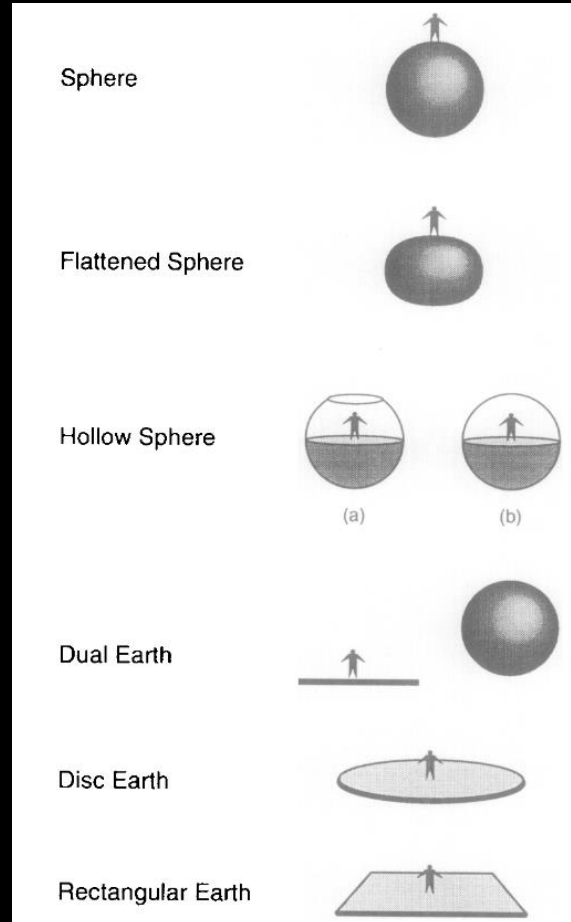


Fig. 1. Item 7 of the EARTH-2: What happens when you walk along a straight line for a very long time?

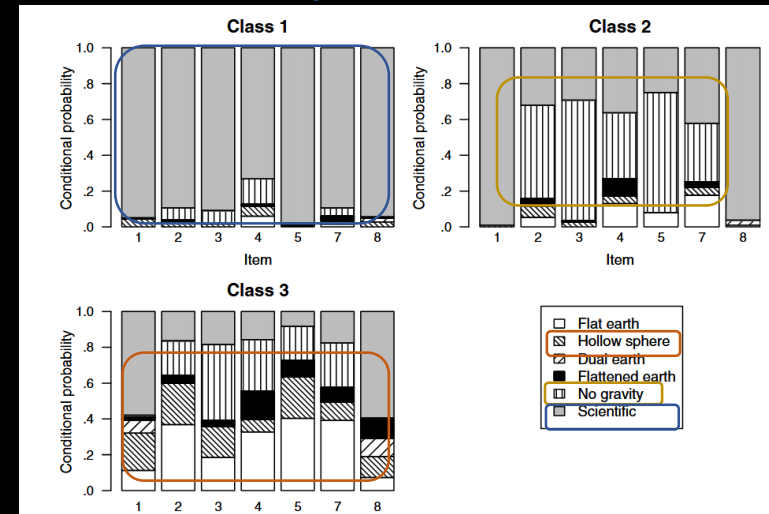
Theory Conceptual Change Model

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Results *Scientific concept*



No gravity-conception

No gravity-/hollow sphere-conception

- Overall: Little evidence for coherent mental models, but also for fragmentation

Theory Conceptual Change Model

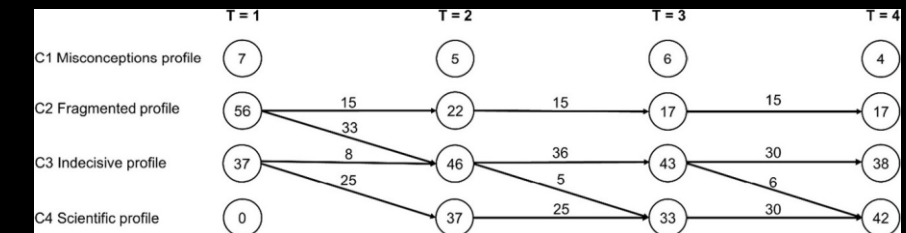
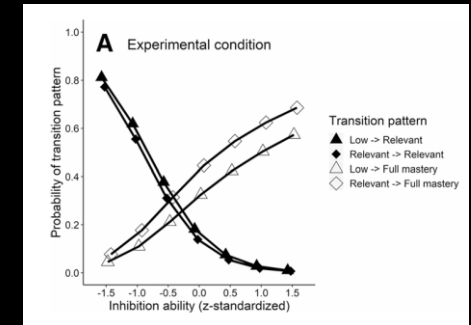
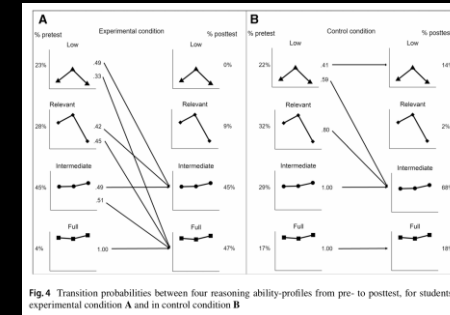
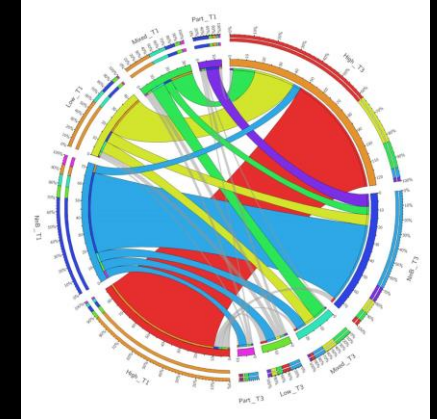
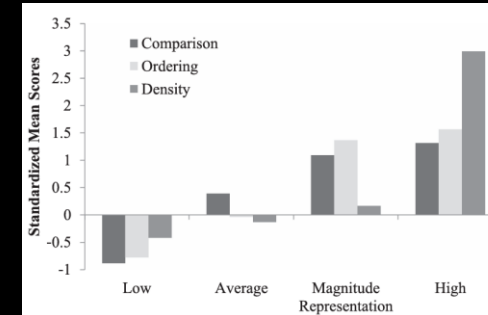
Following up on Schneider & Hardy (2013)

Theoretical assumptions about:

- Concepts about rational numbers
- Training hypothetico-deductive reasoning in children
- Higher education students' concepts of human memory

McMullen et al. (2015)
Edelsbrunner et al (2018)
Grimm et al. (2023)
Kainulainen et al. (2017)

...



Theory Conceptual Change Model

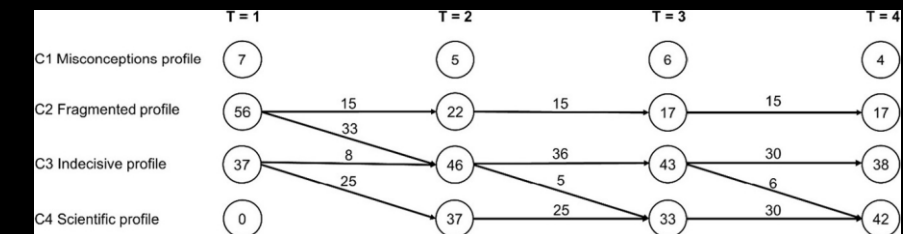
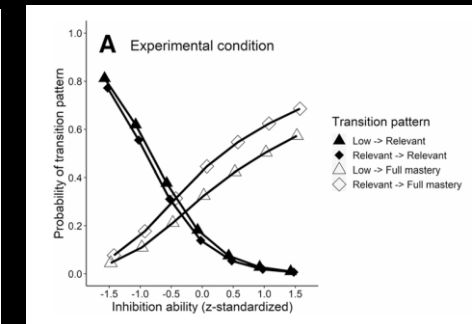
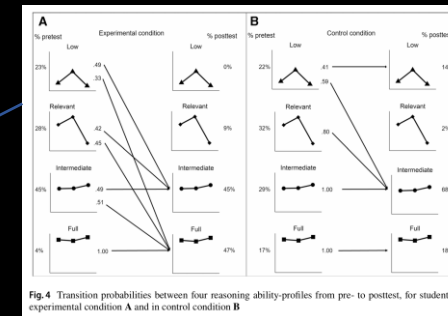
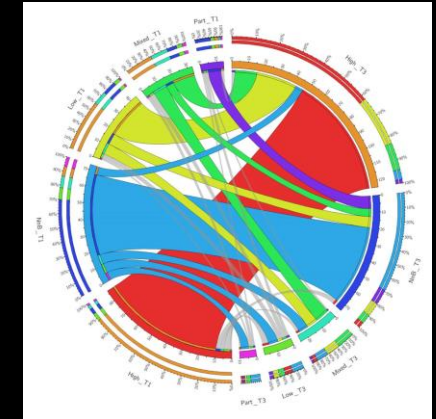
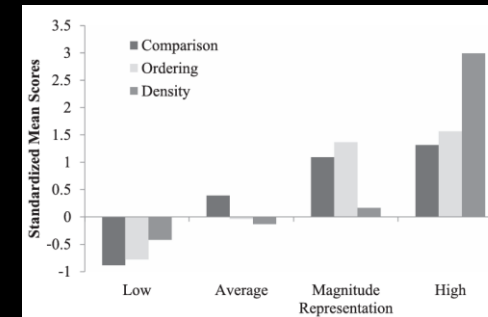
Following up on Schneider & Hardy (2013)

Novel theoretical insights:

- Knowledge of magnitude representations is necessary, but not sufficient, for knowledge of density concepts
- Inhibition predicts acquisition of concepts about insufficient evidence
- Knowledge integration occurs during complex concept acquisition

McMullen et al. (2015)
Edelsbrunner et al (2018)
Grimm et al. (2023)
Kainulainen et al. (2017)

...



Theory Conceptual Change Model

Historical observations:

- Move away from theory-driven (i.e., confirmatory) modeling **to data-driven** (i.e., exploratory) modeling
- Integrations of educational theory into statistical models centers around **qualitative individual differences** (e.g., stages, strategies, schemas, misconceptions)
- These individual differences are represented using **mixture modeling**
- Mixture models allowed **theory testing** and supported **theory refinement**

Theory

Model

Part I

Theory integration in
statistical models

Model integration
into educational
research

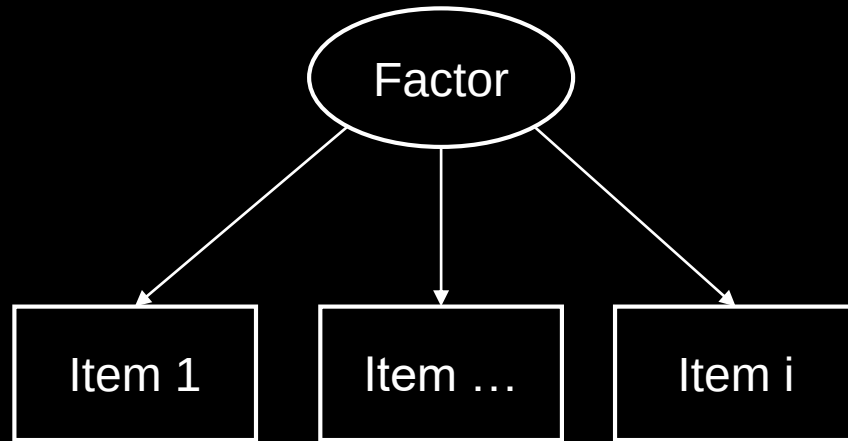
Part II

Part III

Beyond traditional
theory-model
ontologies

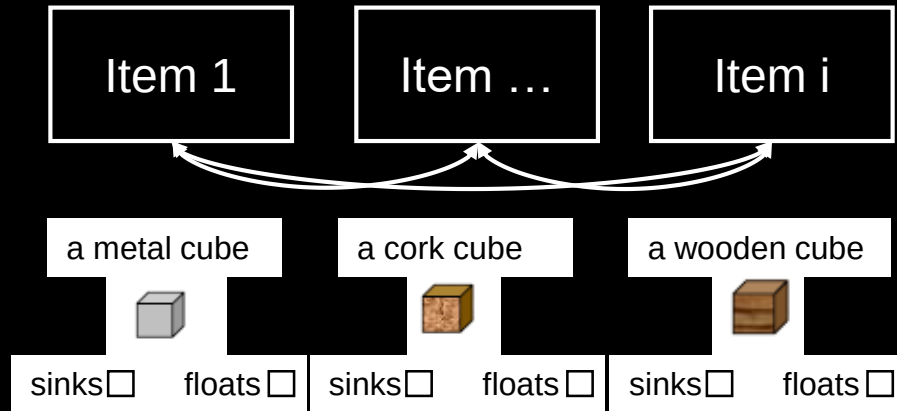


Model



Theory

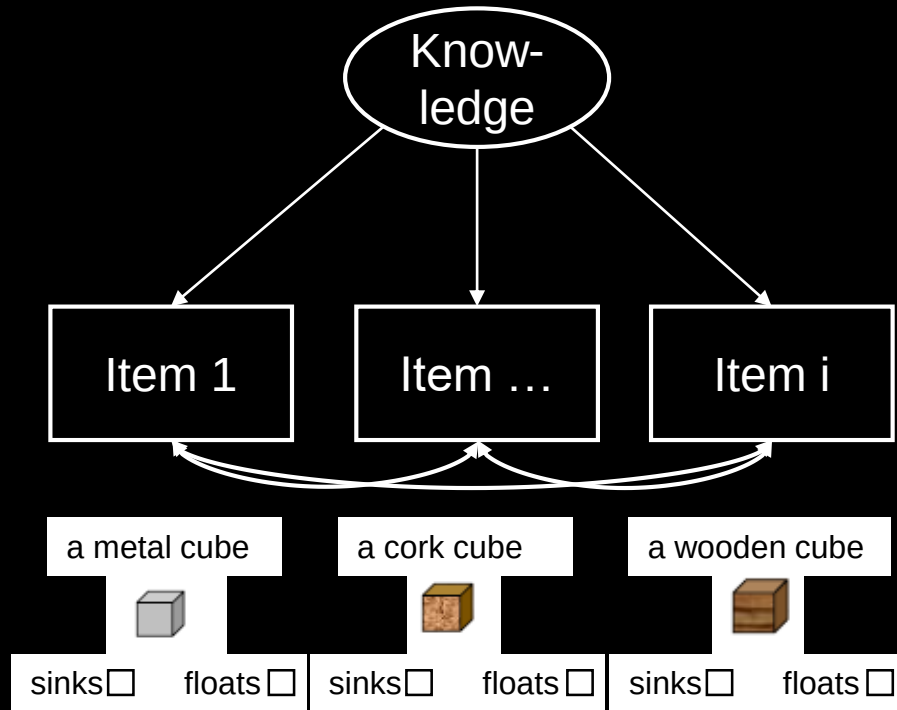
Model



Responses to items **correlate** with one another

Theory

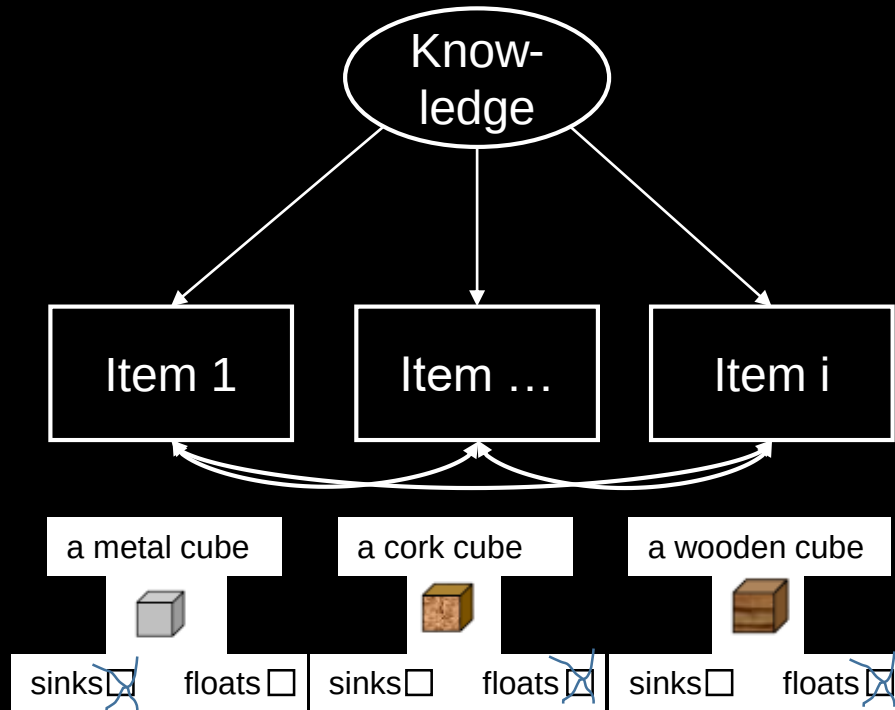
Model



Responses to items **correlate** with one another
Because of a common underlying **factor** –
Children`s **knowledge**

Theory

Model



Responses to items **correlate** with one another
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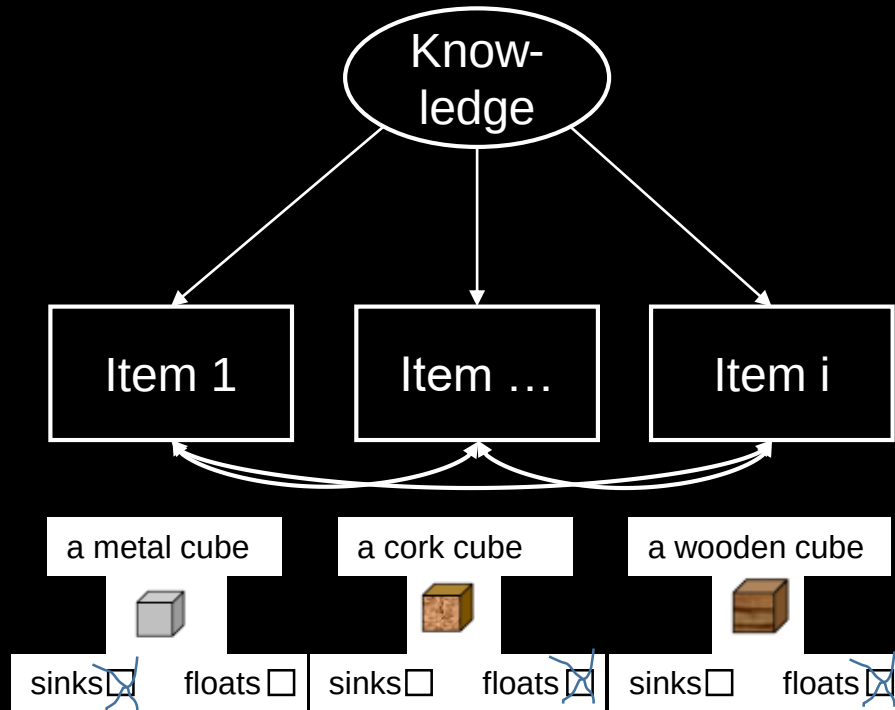
Theory

*If we assume that all items measure knowledge about floating and sinking, **then** a factor model postulating one common factor must fit to our data*

90s:
*Rule of factor
analysis ritual*

- (1) Develop (many) items
- (2) Fit a (exploratory or confirmatory) factor analysis model
- (3) Compare results to theory
- (4) Remove items that do not fit model

Model



Responses to items **correlate** with one another
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Theory

Epistemic beliefs about knowledge, its sources, and development (Perry, 1970)

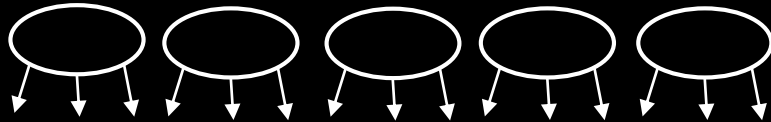
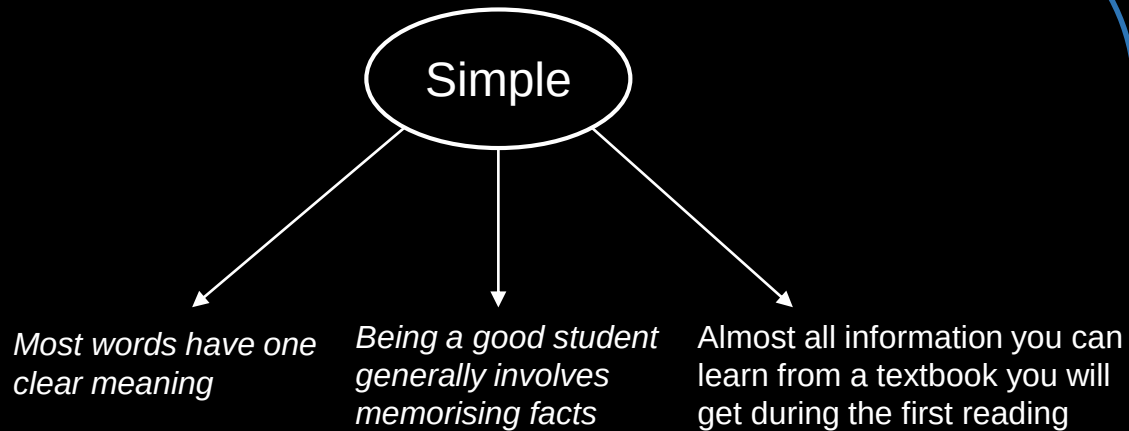
Beliefed to affect learning strategies and outcomes

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Model

Schommer (1990, 1993; 1994)



Source: Omniscient authority <-> reasoned out
Certainty: Absolute <-> evolving.
Organization: Compartmentalized <-> integrated
Control: determined <-> acquired
Speed: quick <-> gradual

Theory

Epistemic beliefs about knowledge and knowing
(Perry, 1970)

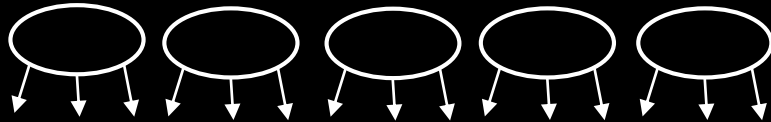
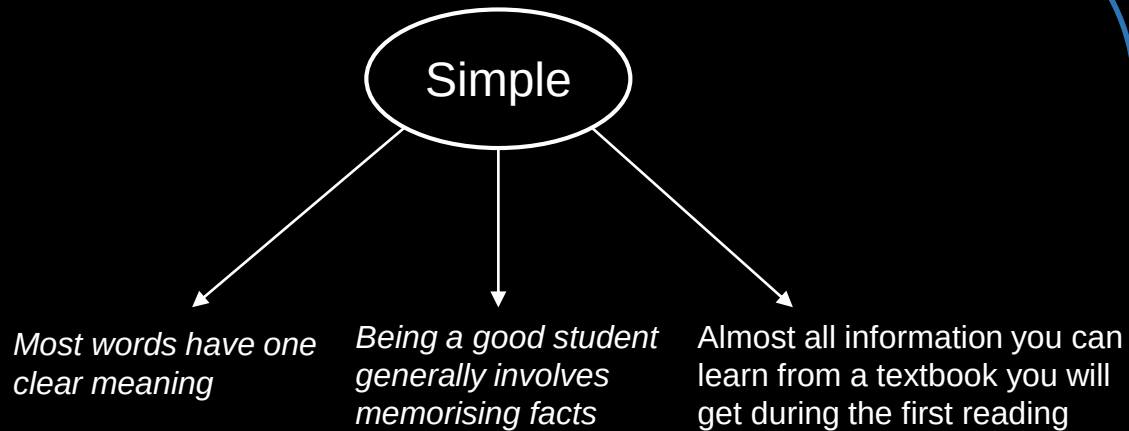
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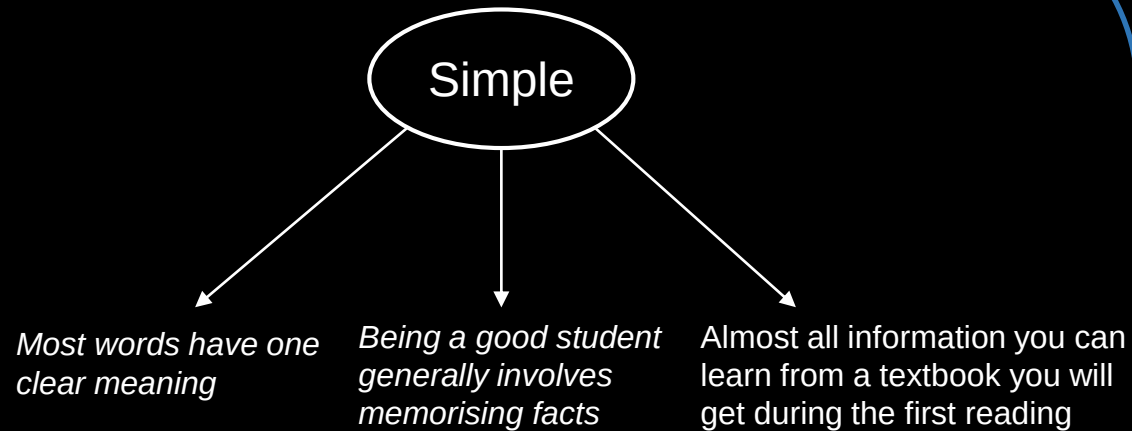
Theory

- Multiple new developments & revisions of epistemic belief questionnaires (Greene et al., 2010)
- Refinement of postulated dimensions and items (e.g., Chan & Elliott, 2004; Chai et al., 2006; Conley et al., 2004; Schiefer et al., 2022)

90s – 2000s:
**The rise of
epistemic beliefs**

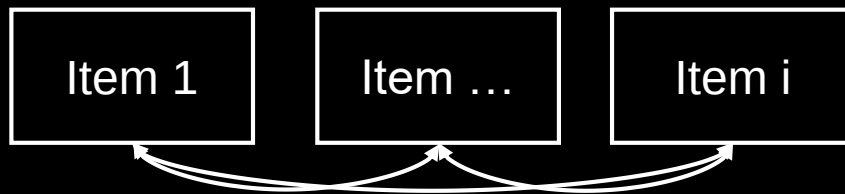
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Model



- Move towards **epistemic cognition in action** because *you cannot measure what is not stable*

- If responses do **not correlate** between items, you should not regard them as a **unitary construct**



Theory

- Criticisms of instable factor structures across samples, low internal consistencies (Cronbach's Alpha; DeBacker et al., 2008), contextual dependence (Merk et al., 2018), interpretational multiplicity (Greene & Seung, 2014)

- Handbook of Epistemic Cognition: **Nail in the coffin of epistemic beliefs research** (e.g., Mason, 2016)

2010s: The downfall of epistemic beliefs

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Theory

Model

Part I

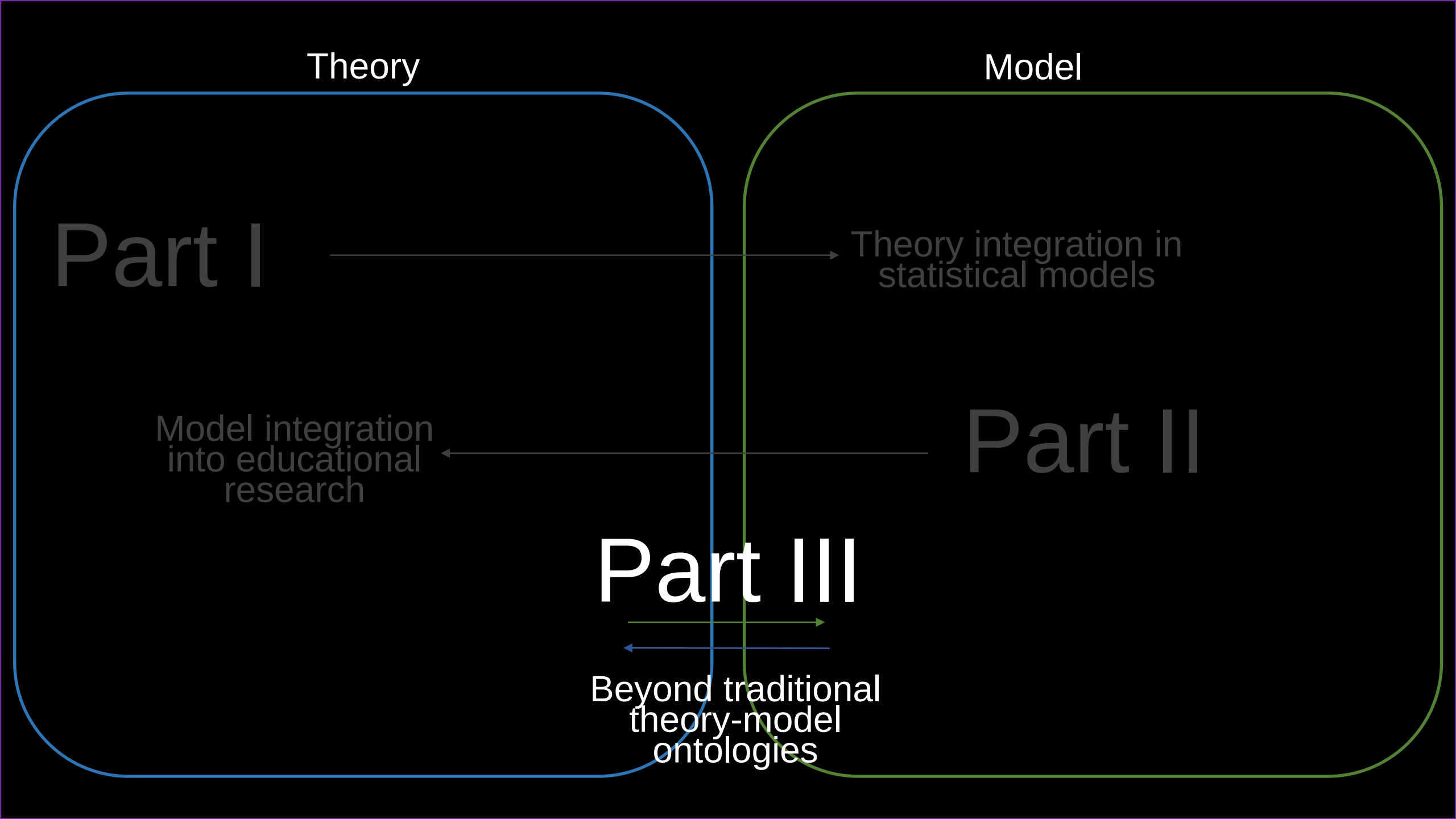
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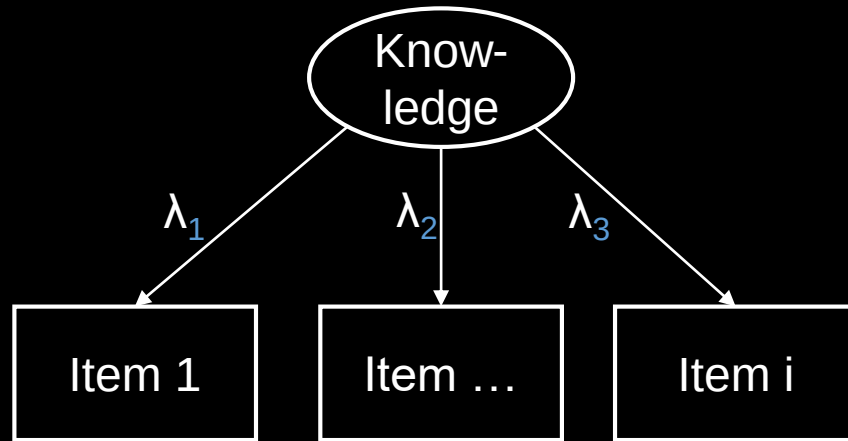
Part II

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Beyond traditional
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Model



Since 1990s:

The factor structure (factor loadings etc.) must be the same across samples/ time points to be able to compare means, correlations, predictive strength (Meredith, 1993)

Measurement invariance

Theory

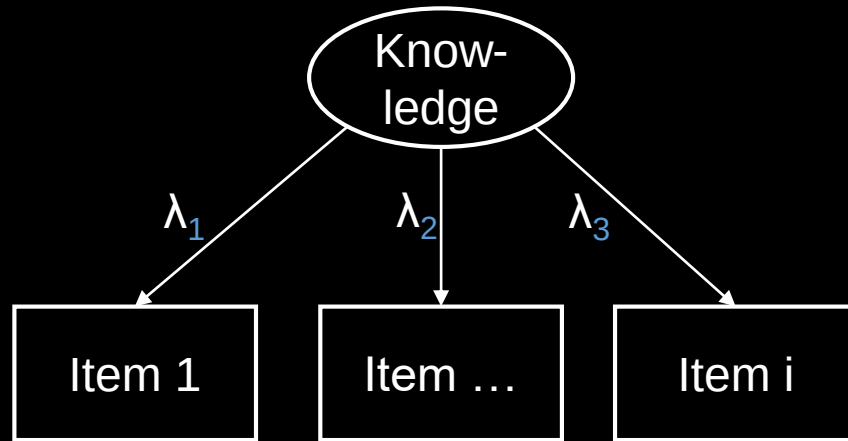
Since 2010s

Questionnaire/test data often thrown away if this statistical assumption is not given

Extreme case:
Epistemic beliefs (construct)

Wide-spread criticism of studies not investigating MI (Maasen et al., 2025)

Model



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Measurement invariance

Theory

Alternative view

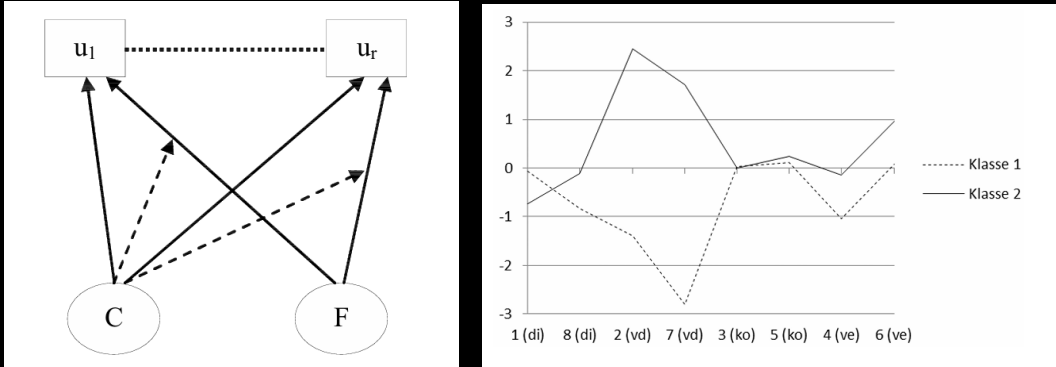
MI is not a prerequisite for any comparisons (Robitzsch & Lüdtke, 2023) because...

- MI modeling decisions are arbitrary (Robitzsch & Lüdtke, 2023)
- MI models error that we do not want in our model (cf. Boone & Staver, 2020)

«Violation» of MI can be...

- Expected (cf. Conceptual change)
- Interesting

Model



Example: Kleickmann (2011)

Factor-mixture modeling

- Violation of MI becomes a **feature**
- Groups with measurement differences represent **knowledge restructuring**
- Children with same measurement properties regarding density, condensation, evaporation **differ in displacement** (cf. Wilson, 1993)

Theory

Alternative view

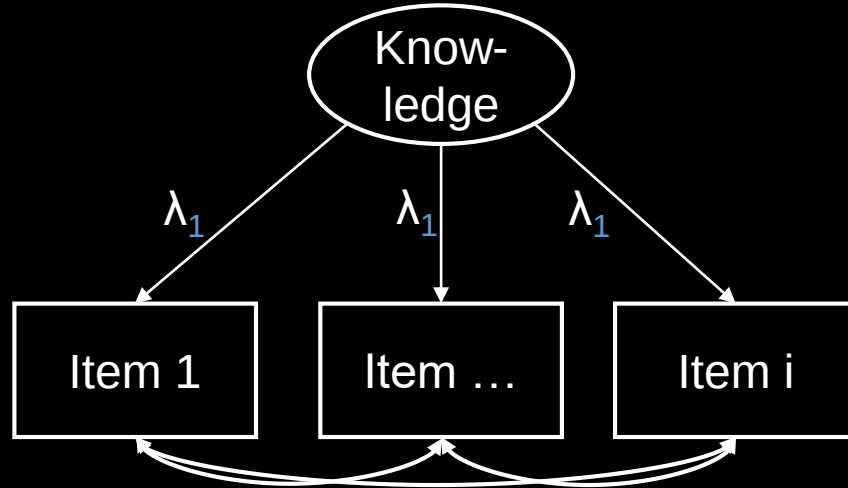
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- MI modeling decisions are arbitrary (Robitzsch & Lüdtke, 2023)
- MI models error that we do not want in our model (cf. Boone & Staver, 2020)
- Analogy to moderation analysis:/causal inference A measurement model across groups is a marginal/average model

«Violation» of MI can be...

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Model



The sum score is a *constrained version* of factor analysis
(McNeish & Wolf, 2020)

- All items are *equally weighted*
- Represents the implicit assumption of *equal factor loadings*

because

Theory

Bullshit!

OFC!

Yes!

Bullshit!

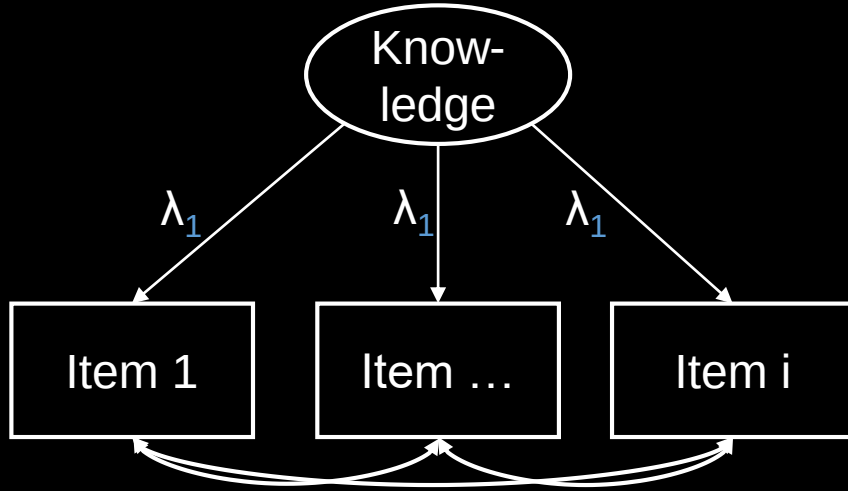
0. Sum/mean scores make the assumption of equal factor loadings

1. This assumption is testable via factor analysis

2. Empirically, this assumption rarely holds

3. Conclusion:
We should not use sum scores/means at all

Model



Theory

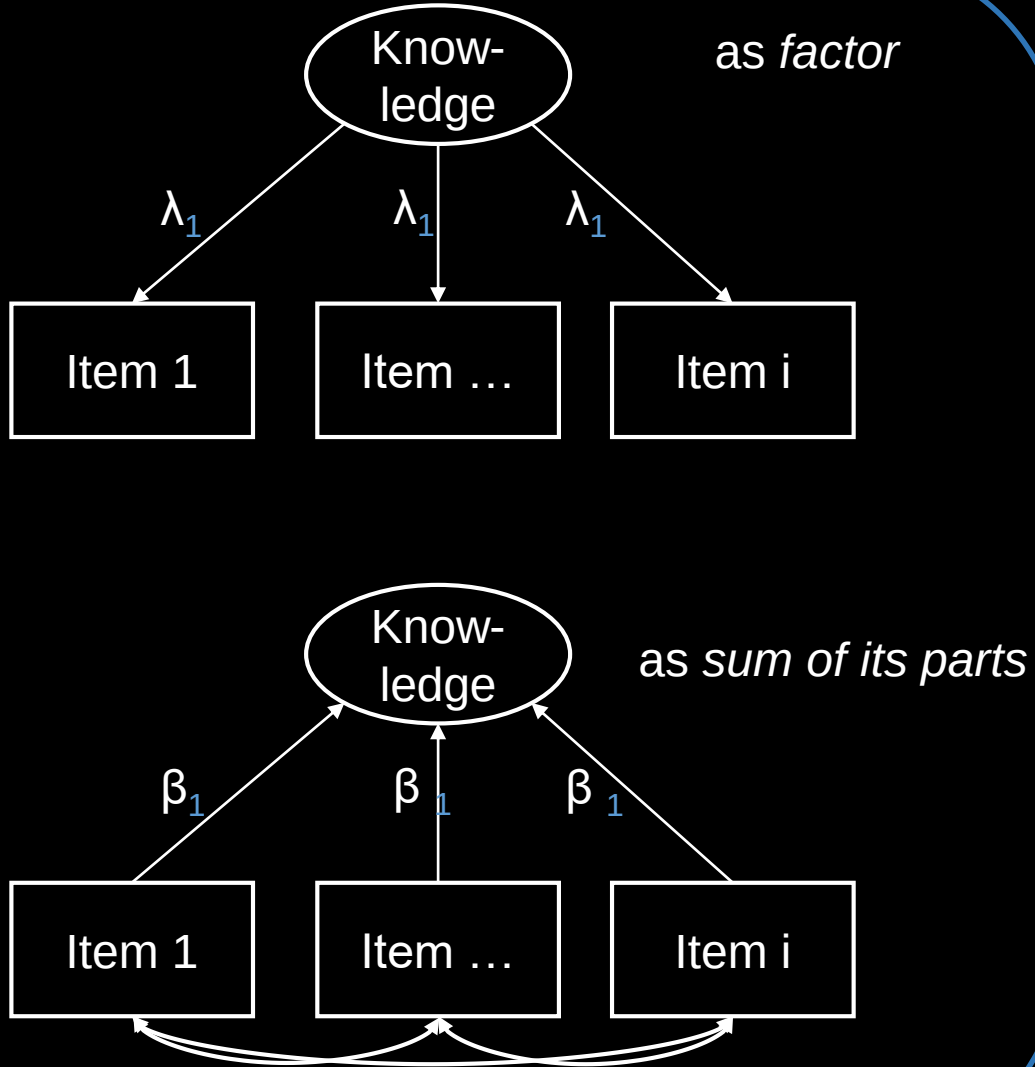
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Theory

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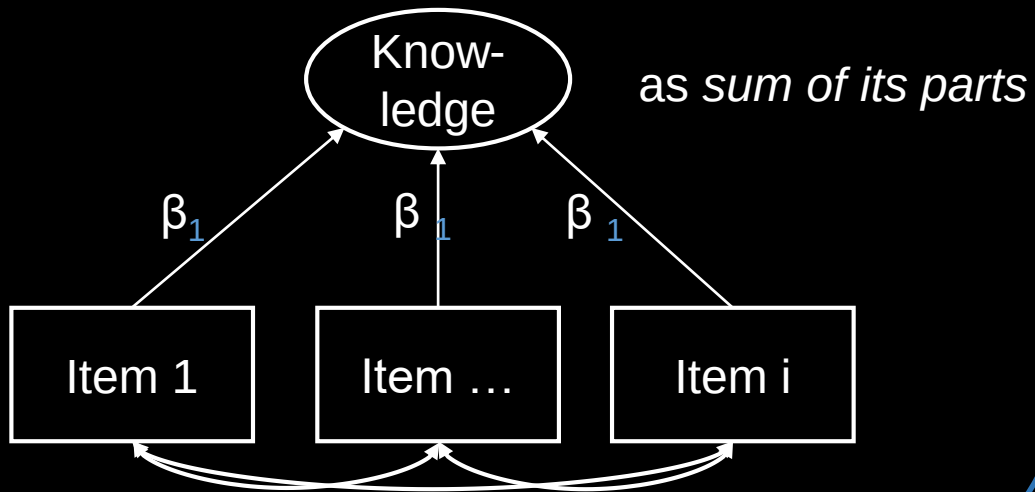
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We should not use sum scores/means at all

Model

Knowledge can be considered a **formative or composite construct** (Taber, 2018; Stadler et al., 2021; Edelsbrunner, 2022; Edelsbrunner et al., 2025)

We can use it as a **functionally unidimensional construct** (Baird et al., 2017)

No need for interitem correlations or high Cronbach's Alphas (Edelsbrunner et al., 2025)



Theory

0. Sum/mean scores make the assumption of equal factor loadings

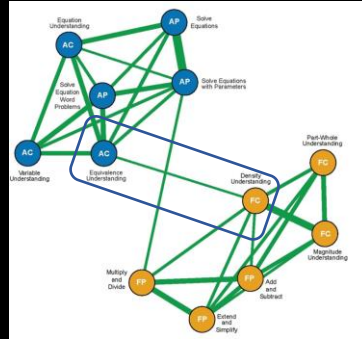
1. This assumption is testable via factor analysis

2. Empirically, this assumption rarely holds

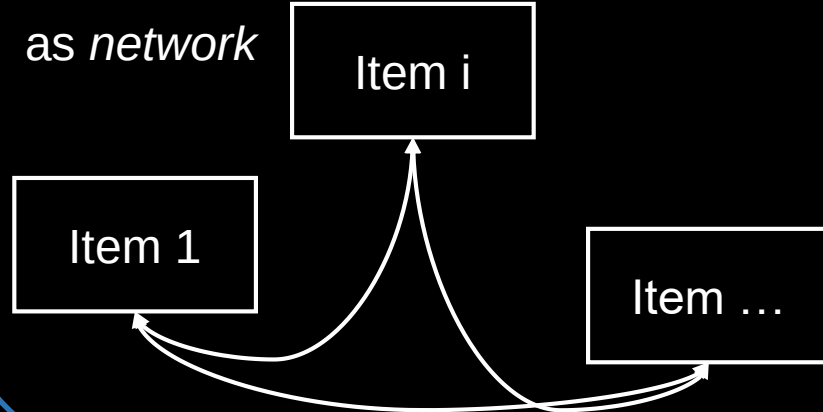
3. Conclusion:
We should not use sum scores/means at all

Model

- **Mutual dependencies/ developmental effects between items** (van der Maas et al., 2006; vanderWeele & Batty, 2023)
- **No latent variable**
- Unique (i.e., partial) correlations between items point to **mutual dependencies/influences**



Example: Forsmann et al. (2026)
 Fraction density $\leftrightarrow$ algebraic equivalence
as network



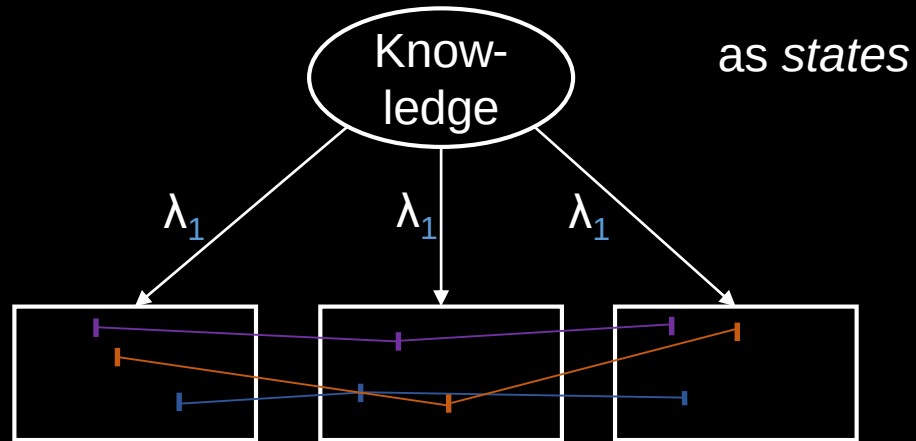
Theory

If responses do **not correlate** between items, you should not regard them as a **unitary construct**

0. Sum/mean scores make the assumption of equal factor loadings

We can choose from different ontologies (meta-theories) and theories to describe our construct of interest

Model



e.g., Kleickmann, 2011; Schneider & Hardy, 2013; Tasuoka, 1985; Wilson, 1989

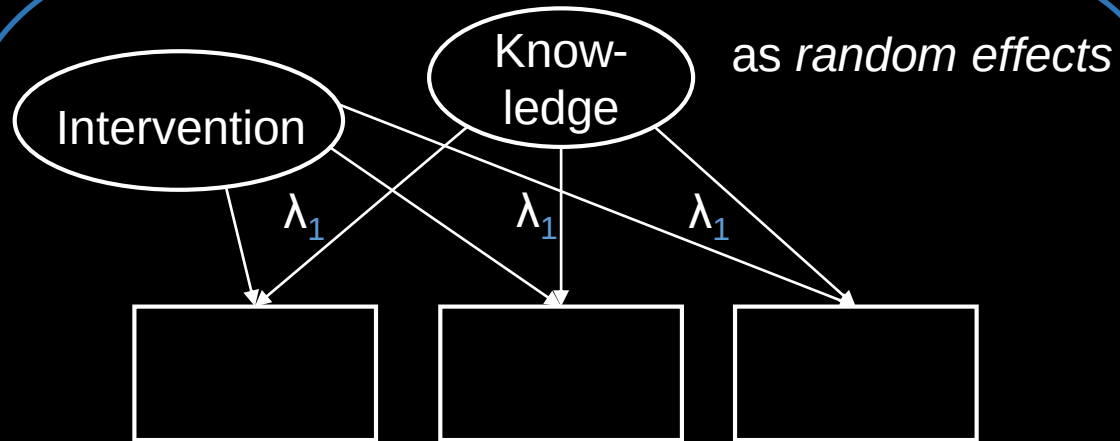
Theory

If responses do **not correlate** between items, you should not regard them as a **unitary construct**

0. Sum/mean scores make the assumption of equal factor loadings

We can choose from different ontologies (meta-theories) and theories to describe our construct of interest

Model



Multilevel modeling

- Items are treated as construct-relevant contexts (Donnellan et al., 2023)
- Effects of interventions or development may systematically **differ** across items
- Allows theoretical **generalization** across items (Yarkoni, 2020)

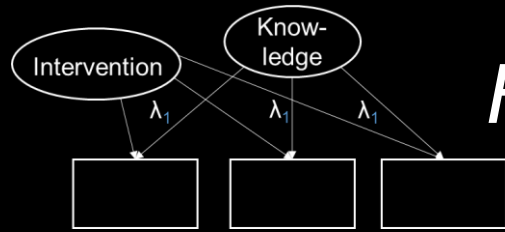
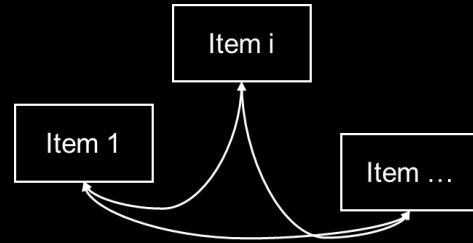
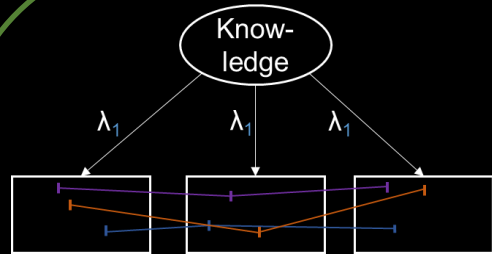
Theory

If responses do **not correlate** between items, you should not regard them as a **unitary construct**

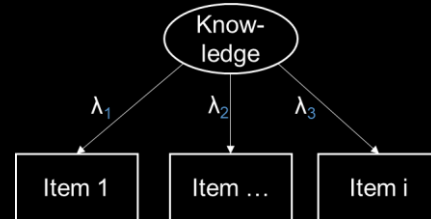
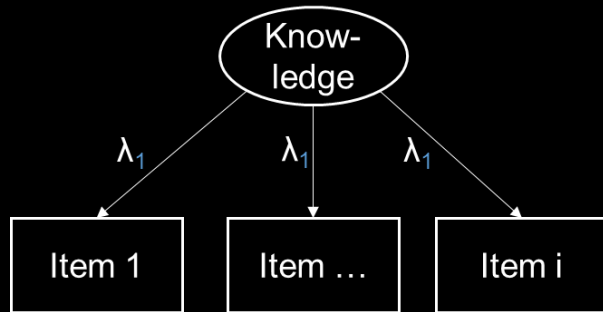
0. Sum/mean scores make the assumption of equal factor loadings

We can choose from different ontologies (meta-theories) and theories to describe our construct of interest

Model



How?



Theory

If responses do **not correlate** between items, you should not regard them as a **unitary construct**

0. Sum/mean scores make the assumption of equal factor loadings

Group/time comparisons require measurement invariance

We can choose from different ontologies (meta-theories) and theories to describe our construct of interest

Model

Option 1: The **empirical** route

Statistical **difference tests**
between ontologies...

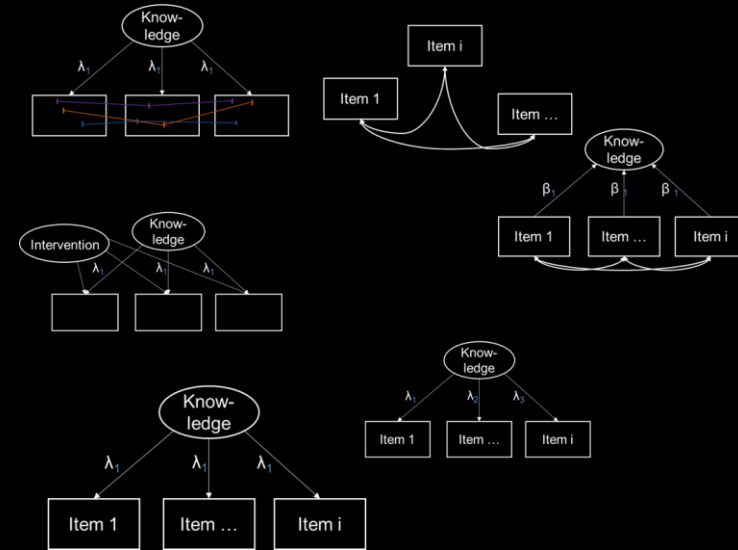
Factor vs. Mixture (e.g., DeBoeck & Wilson, 2005)

Factor vs. network (e.g., van Bork et al., 2021)

Factor vs. composite (e.g., Siegel et al., 2026)

...and theories (e.g., McNeish & Wolf, 2020)

Theory



We can choose from different ontologies (meta-theories) and theories to describe our construct of interest

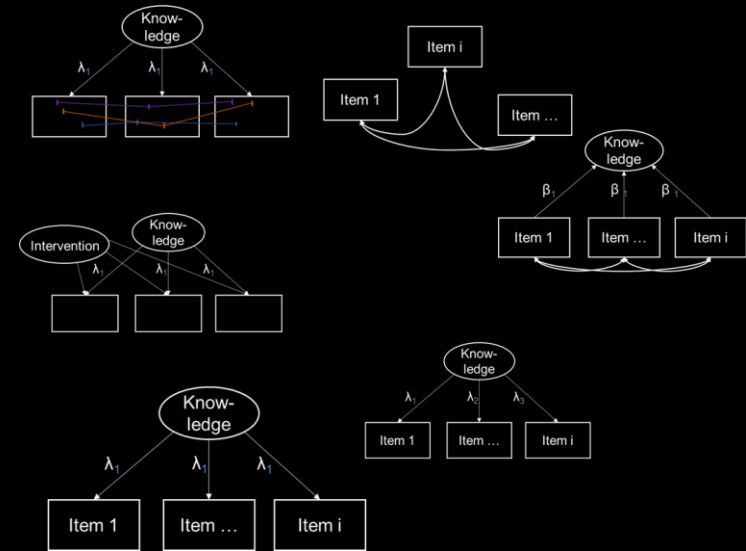
Model

Option 2: The **theoretical** route

Step 1: Decide on your **ontology**
Factor vs. mixture vs. network...

Step 2: Decide on your **theory**
General vs. item-specific effects

Theory



We can choose from different ontologies (meta-theories) and theories to describe our construct of interest

Model

Theory

One of these is **difficult**...

Option 1: The **empirical** route

Statistical **difference tests**
between ontologies...

Factor vs. Mixture (e.g., DeBoeck & Wilson, 2005)

Factor vs. network (e.g., van Bork et al., 2021)

Factor vs. composite (e.g., Siegel et al., 2026)

...and theories (e.g., McNeish & Wolf, 2020)

Option 2: The **theoretical** route

Step 1: Decide on your **ontology**
Factor vs. mixture vs. network...

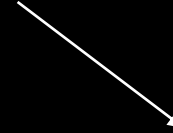
Step 2: Decide on your **theory**
General vs. item-specific effects

One of these is **futile**...

Model

Theory

One of these is **difficult**...



Option 1: The **empirical** route

Statistical **difference tests**
between ontologies...

Factor vs. Mixture (e.g., DeBoeck & Wilson, 2005)

Factor vs. network (e.g., van Bork et al., 2021)

Factor vs. composite (e.g., Siegel et al., 2026)

...and theories (e.g., McNeish & Wolf, 2020)



Option 2: The **theoretical** route

Step 1: Decide on your **ontology**
Factor vs. mixture vs. network...

Step 2: Decide on your **theory**
General vs. item-specific effects

One of these is **futile**...

Model

Theory

Beware of the **inverse fallacy** *fit $\neq$ theory* (Edelsbrunner, 2022)

“Factor analysis may be useful in generating hypotheses about underlying causally efficacious univariate latent variables, but does nothing to establish this”

(vanderWeele, 2020)

Option 1: The **empirical** route

Statistical **difference tests**
between ontologies...

Factor vs. Mixture (e.g., DeBoeck & Wilson, 2005)

Factor vs. network (e.g., van Bork et al., 2021)

Factor vs. composite (e.g., Siegel et al., 2026)

...and theories (e.g., McNeish & Wolf, 2020)

- Different ontologies typically predict the **same means, variances, covariances** (e.g., Marsman et al., 2018; Loken & Molenaar, 2007)
- Difference tests make unrealistic assumptions (**pure ontologies**)
- Theories are **causal**; assessment data **observational** (not experimental/“causal”); **Hidden causal modeling misattribution**

One of these is **futile**...

Model

Theory

One of these is **difficult**...

Option 2: The **theoretical** route

Step 1: Decide on your ontology

Is your construct...

...continuous (more vs. less; e.g., factor/composite)

...categorical (qualitative differences; e.g., mixture)

...both (e.g., factor-mixture, mixture-network, hybrid SEM)?

Step 2: Decide on your theory

Are your items...

...construct relevant (all contributing; composite/ML)

...construct irrelevant (all capturing the same *thing*; factor)

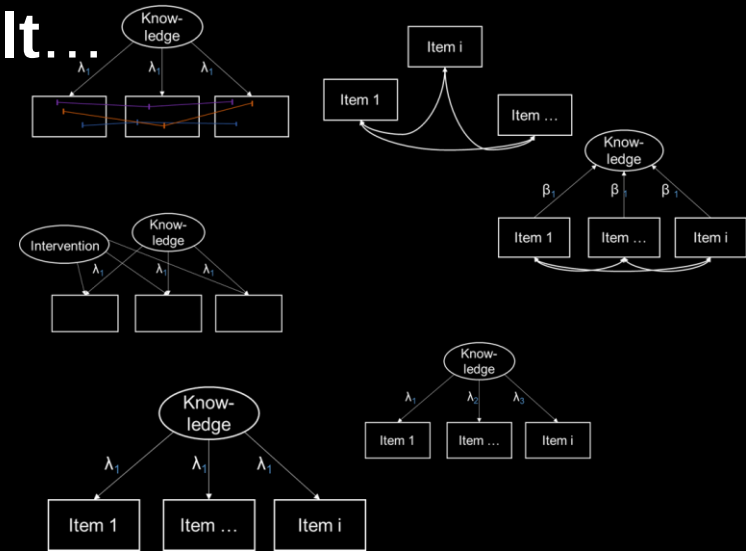
...both (e.g., composite/reflective hybrid SEM)?

Step 3: What do you want?

...equal weights (e.g., sum score, Rasch)

...different weights (e.g., composite, factor)

...both (ML)?



*We can choose from different ontologies (meta-theories) to describe our construct of interest under **theoretical justifications***

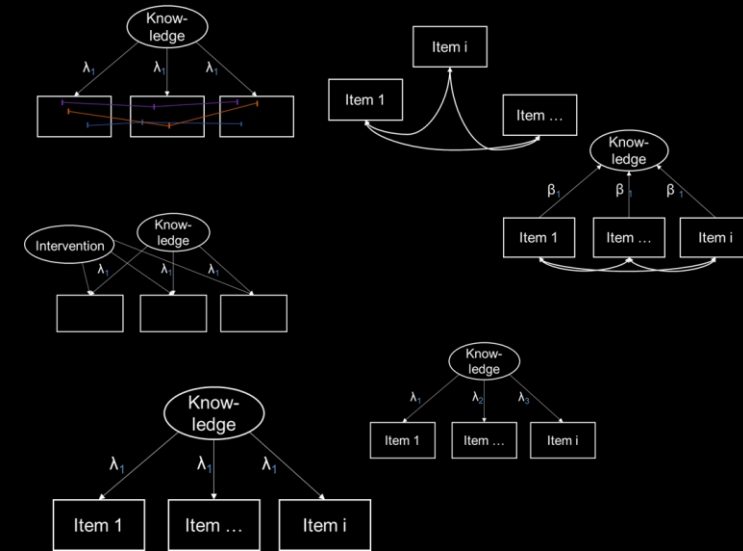
Model

Theory

Option 2: The **theoretical** route

How can we get these?

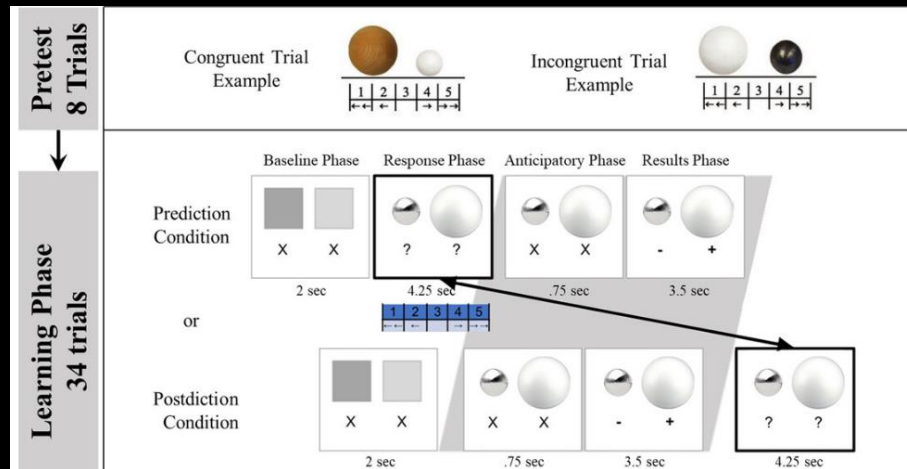
*Theory-informed causal,
formal, cognitive, &
computational modeling*



*We can choose from different
ontologies (meta-theories) to
describe our construct of interest
under **theoretical justifications***

Model

- Colantonio, ... Brod, et al. (2024)
Bayesian computational models of belief updating about water displacement
«volume matters» vs. «material matters»



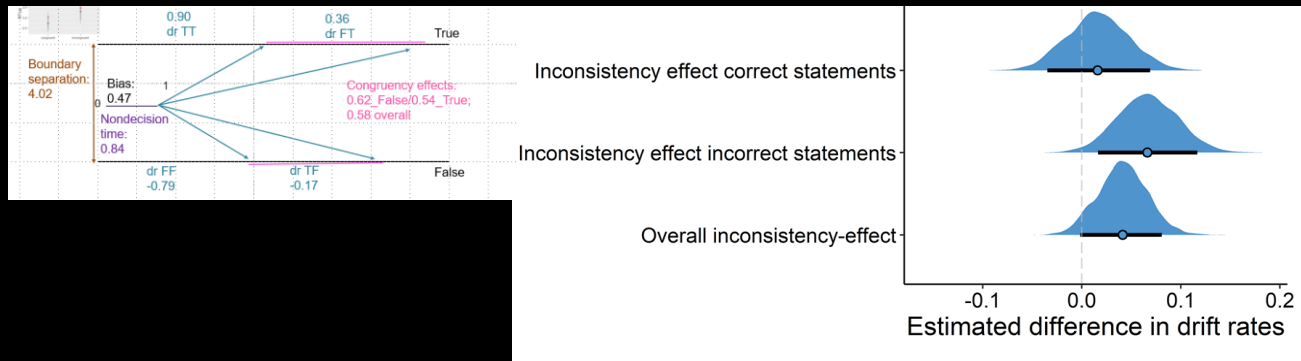
Knowledge tracing showing how
predicting affects **belief revision**

Theory

*We can choose from different ontologies (meta-theories) to describe our construct of interest under **theoretical justifications***

Model

- Edelsbrunner & Singmann (2026)
Wiener Diffusion model about suppressing
intuitive conceptions (Shtulman & Valcarcel, 2012)
«fires provide heat» vs. «coats provide heat»



*We can choose from different ontologies (meta-theories) to describe our construct of interest under **theoretical justifications***

Science expertise affects primarily **correct statements**

Longitudinal design:

No relation with inhibition

Model

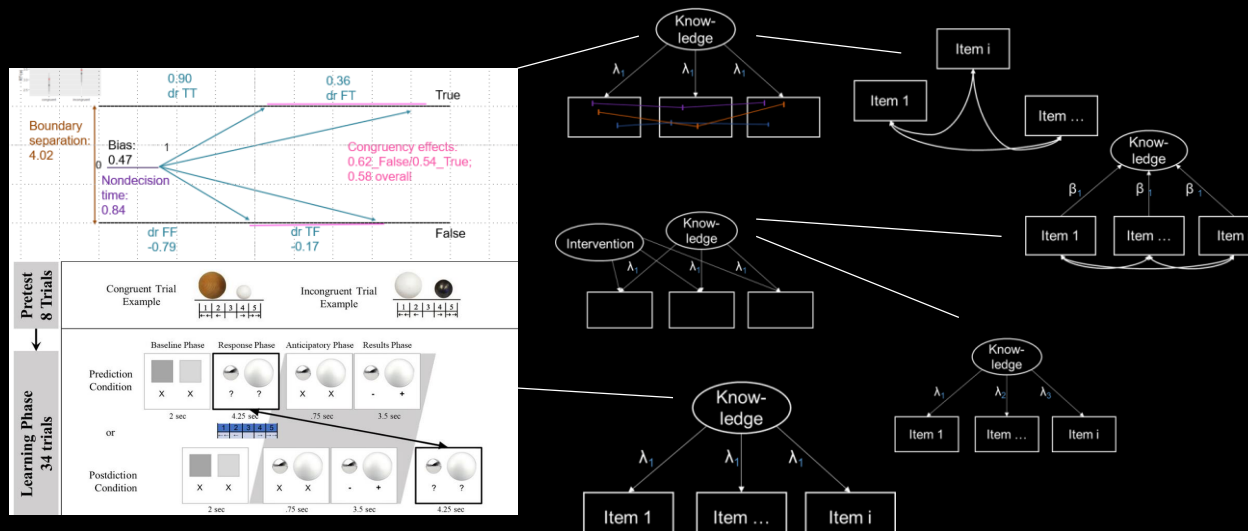
- Back to (semi-) confirmatory modeling (e.g., Wilson, 1989)
- Cognitive modeling
- Computational modeling
- Formal modeling
- Causal inference

Theory

*We can choose from different ontologies (meta-theories) to describe our construct of interest under **theoretical justifications***

Model

- The future is **cognitive, formalized, computational, hierarchical, and causally informed** (Robinaugh et al., 2021)



*We can choose from different ontologies (meta-theories) to describe our construct of interest under **theoretical justifications***

Theory

Specific wishes

- Multilevel modeling for hybrid overall/item-specific effects (Donnellan et al., 2023)
- Cognitive SEM hybrids (e.g., Rouder & Haaf, 2019)
- Data-/GenAI-driven modeling (e.g., observational learning analytics) will be **futile for causal aims** (e.g., personalization)
- Re-thinking of our field as a **cognitive science** that goes from the process of the individual to personalized learning and population development

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thank you.

<https://github.com/peter1328/presentations>